

# Monte Carlo Fundamentals & Motivation for Variance Reduction

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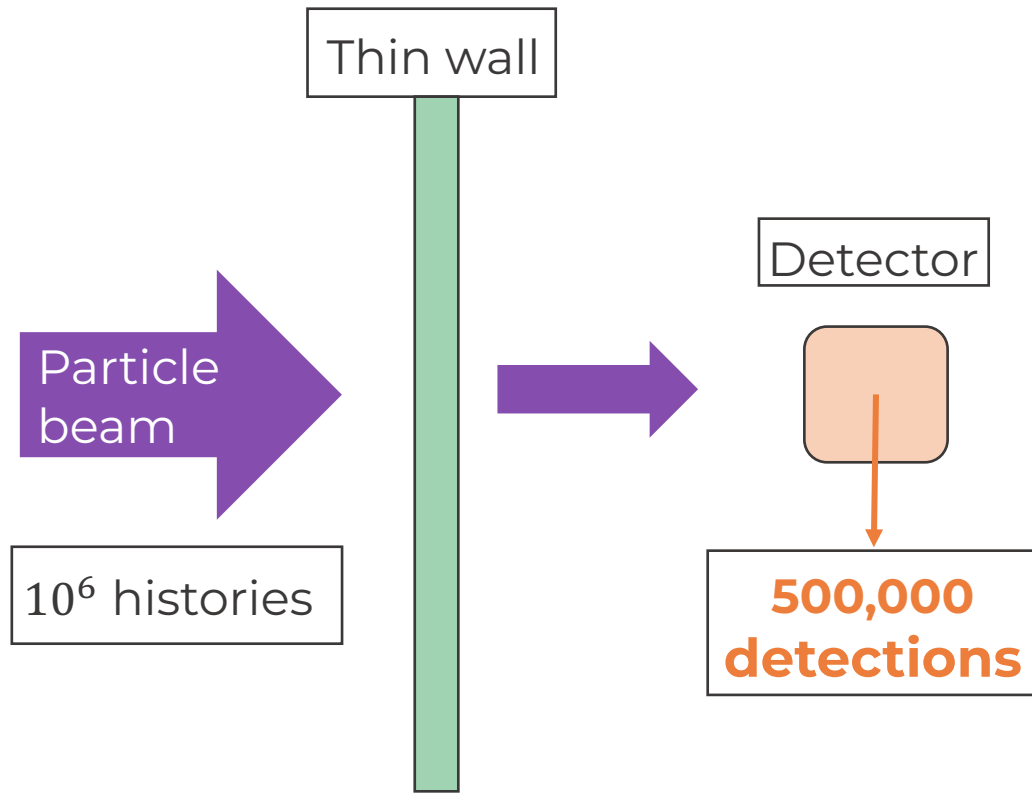
# Outline

1. Motivation
2. Monte Carlo in a Nutshell
3. Why (Sometimes) Brute Force Fails
4. Variance and Efficiency
5. Motivation for Variance Reduction
6. Families of Variance Reduction Techniques
7. Takeaways

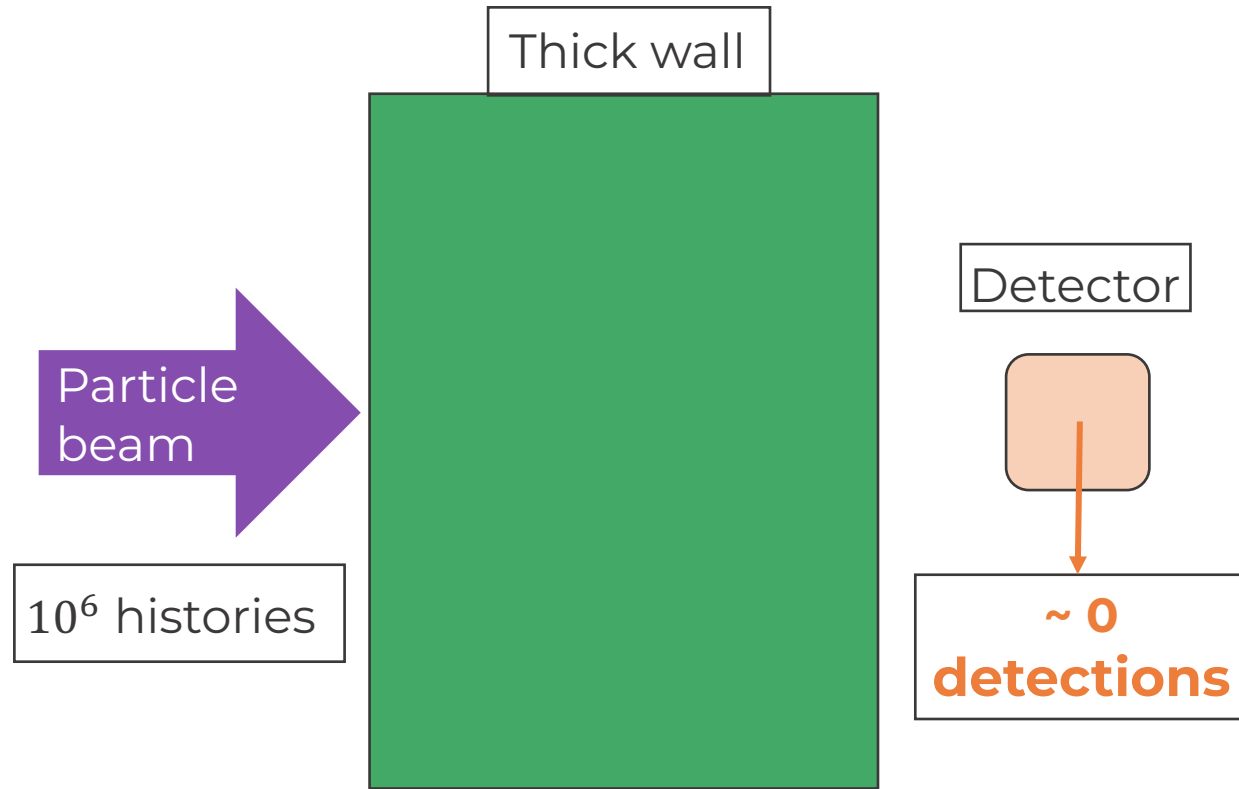
# Motivation

# When Monte Carlo Becomes Statistically Impractical

## The “Easy” Case



## The “Impossible” Case

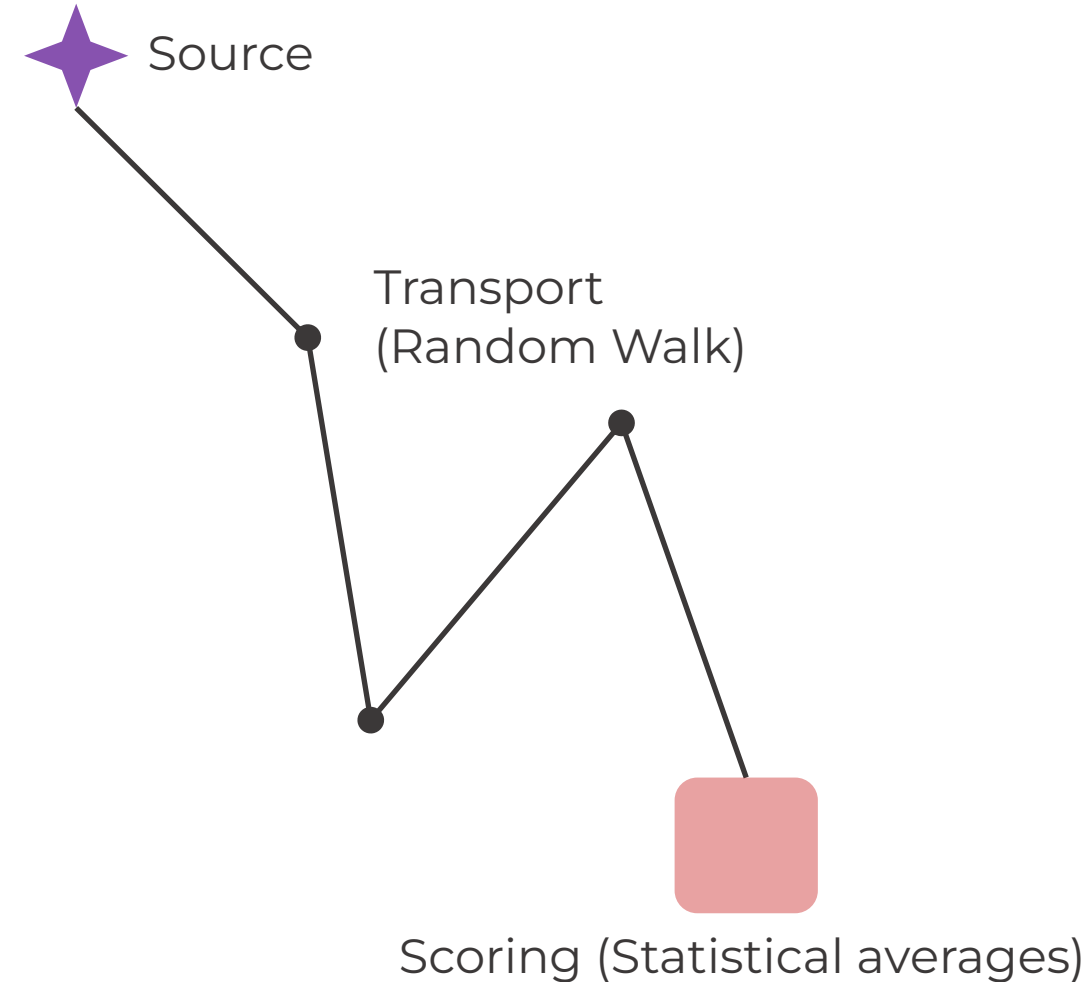


For rare events, simulating reality is **computationally inefficient**.

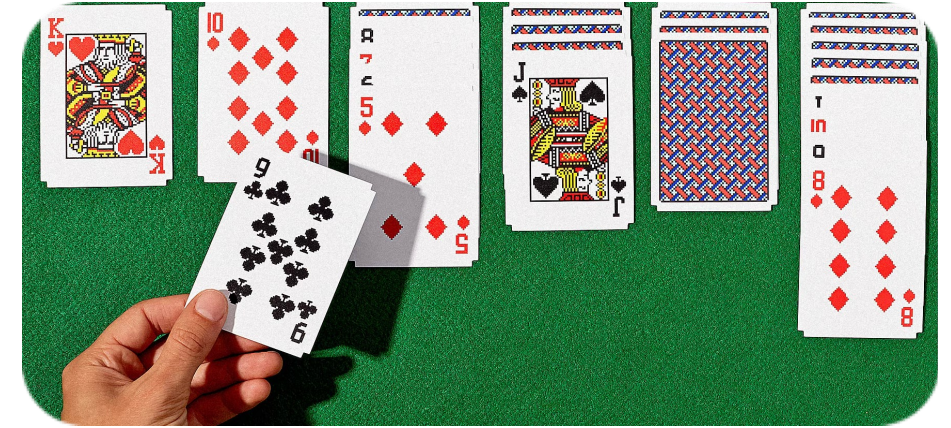
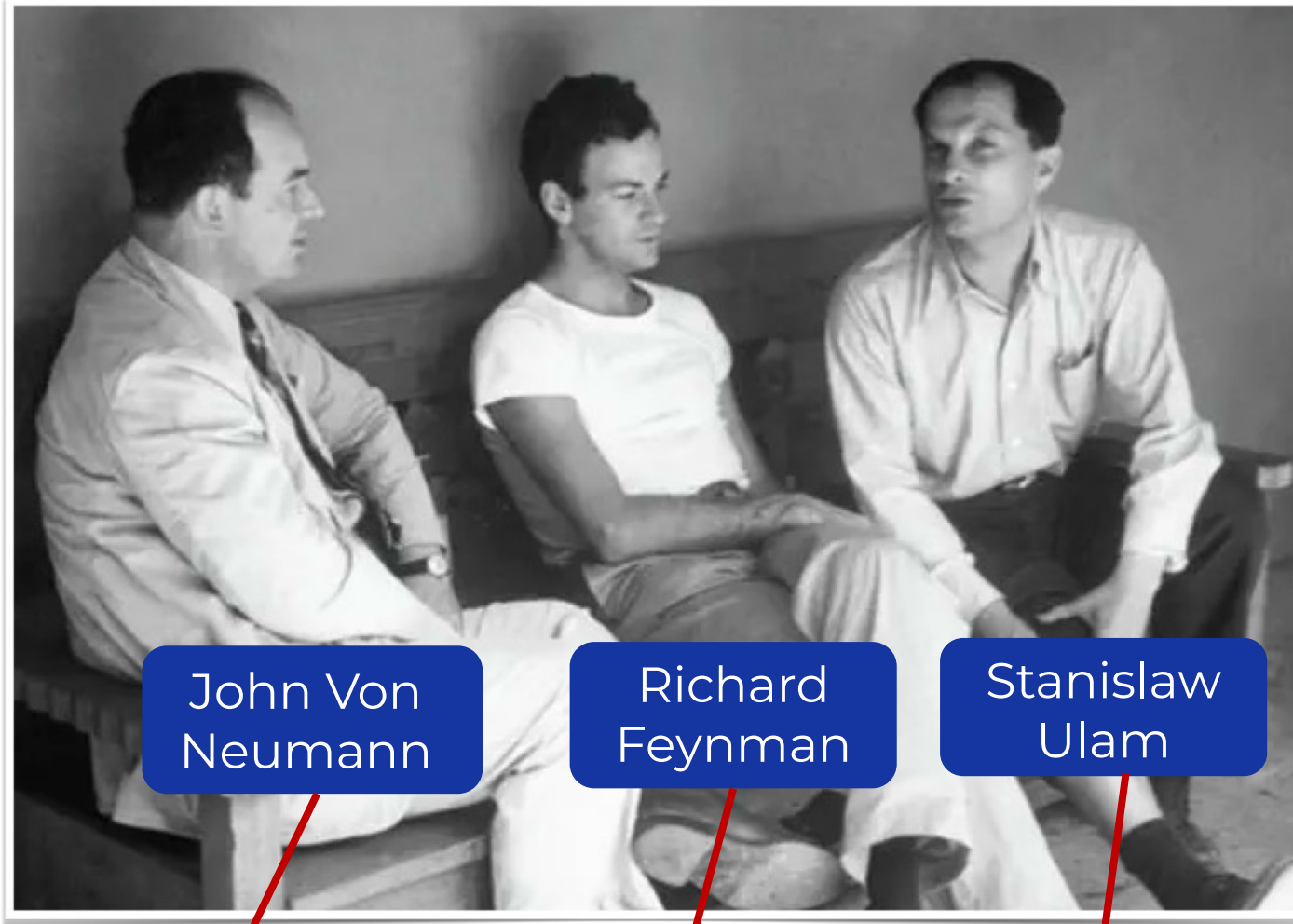
# Monte Carlo in a Nutshell

# What is Transport Monte Carlo?

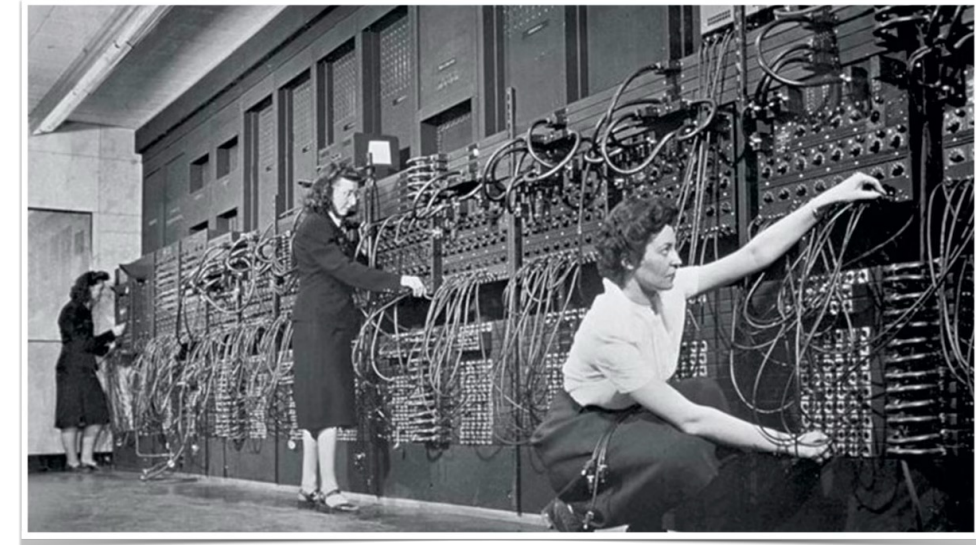
- Numerical experiment: We do not solve the Transport Equation. **We simulate it.**
- Individual Histories: We track millions of **independent particles** from birth to death.
- Physics as probabilities: Physical laws (cross-sections) become **Probability Density Functions** (PDFs).
- The Result: Macroscopic quantities (Flux, Dose, Reaction Rates, Detector Hits...) are **statistical averages** of these histories.



# A little bit of history



Solitaire Game: Stan Ulam's inspiration



ENIAC Computer

The Math

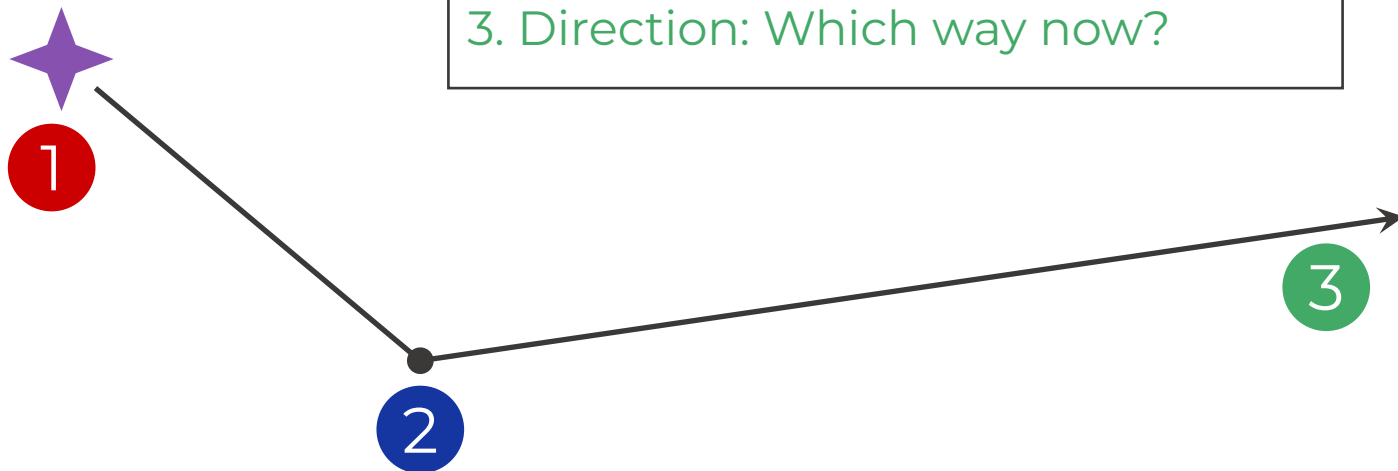
The Physics

The Idea

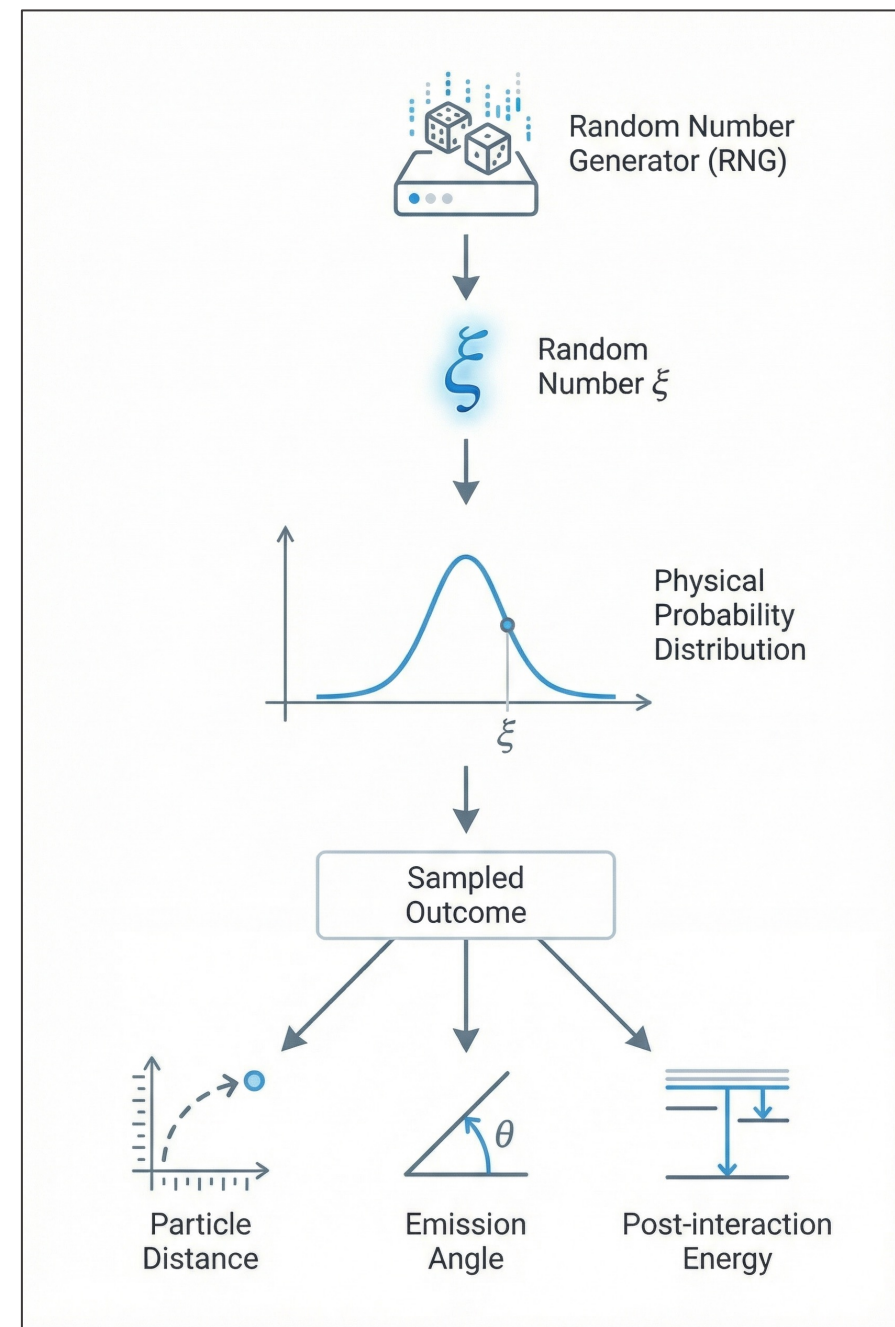


# The Random Walk

1. Distance: How far does it go?
2. Collision: Scattered or absorbed?
3. Direction: Which way now?



- We don't solve equations for these steps.
- We sample (select) outcomes **randomly** based on **physical probabilities** (PDFs).
- Analogy: We roll a die at every step.





# Sampling: Reproducing the Physics

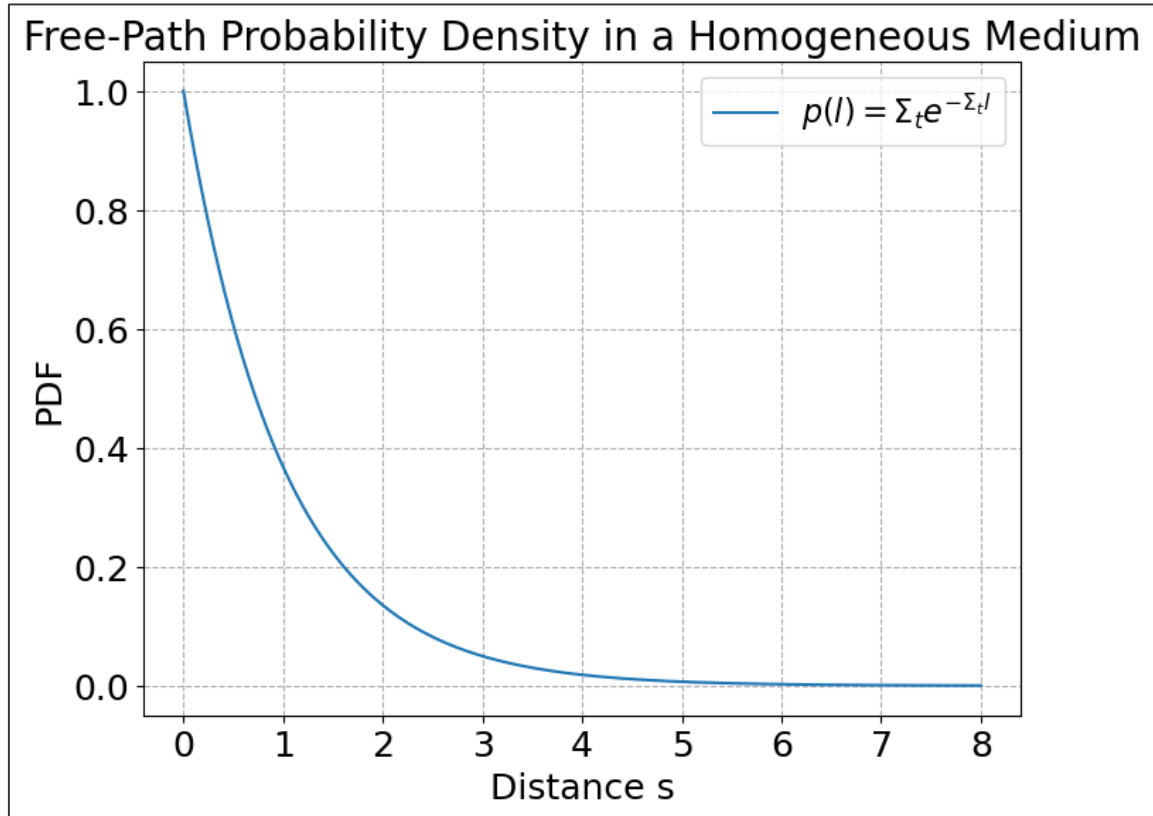
## The Physics

- **Distribution  $f(x)$ :**
  - Nature follows a PDF, e.g.  $f(x) = \Sigma e^{-\Sigma x}$
- **The expected value ( $E[x]$ ):**
  - The physical mean, e.g. Mean Free Path.
  - Given by  $E[x] = \int x \cdot f(x) dx$
  - In our example,  $E[x] = \frac{1}{\Sigma}$

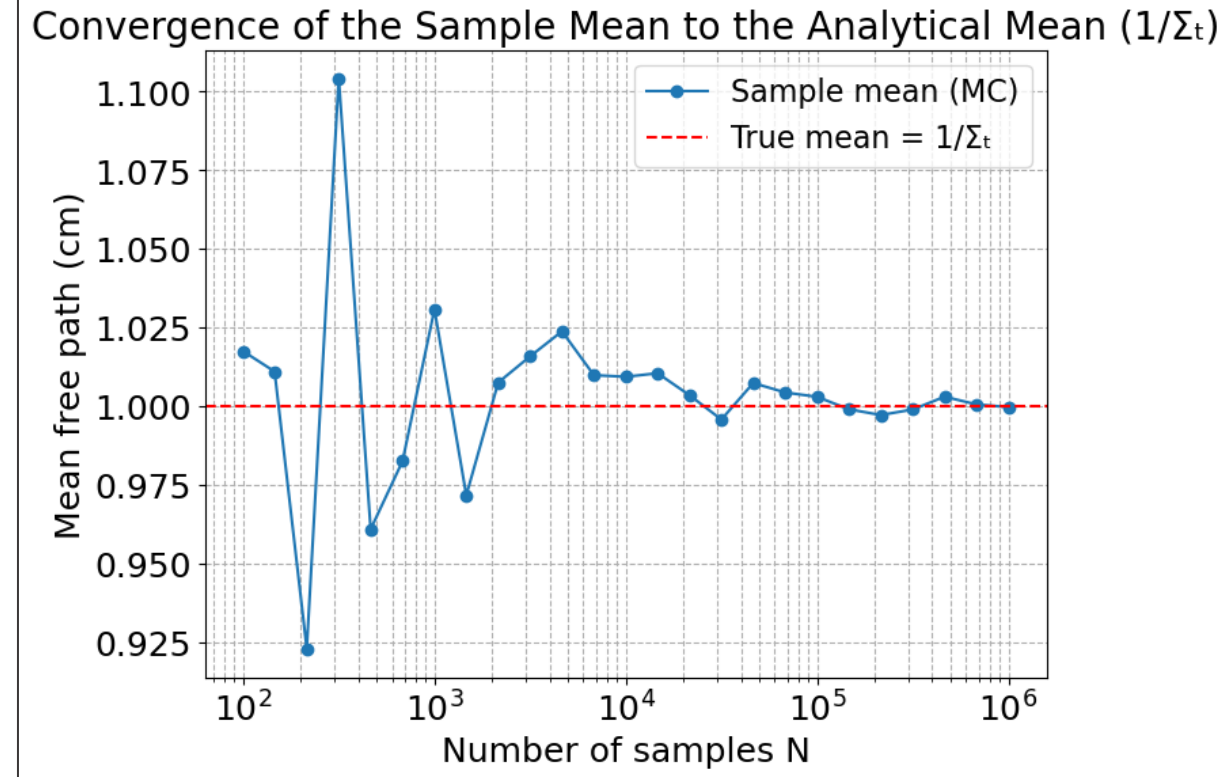
## The Simulation

- **Random Sampling:**
  - Select  $x_i$  values from the PDF using random numbers  $\xi_i$ .
- **Calculate the average of samples:**
  - $\bar{x} = \frac{1}{N} \sum_i x_i$
- As the number of samples  $N \rightarrow \infty$ ,  $\bar{x}$  converges to  $E[x]$  (Law of Large Numbers):
  - $\lim_{N \rightarrow \infty} \bar{x} = E[x]$

# Example: Sample mean vs Analytical Mean



Probability Density Function for the distance a particle travels in an homogeneous medium



Comparison of the sample mean with the “physical mean” for different number of samples

# The General Estimator: $Q = E[h(x)]$

- **Physics Term  $f(x)$ :**

- The probability density of a history  $x$
- **Determined by physics** (cross sections, geometry, boundary conditions, etc.)

- **Score function  $h(x)$ :**

- The value assigned to each history.
- **Defined by the user (scoring)**

- **The result:**

- $$\bar{Q} = \frac{1}{N} \sum_{i=1}^N h(x_i) \approx \int h(x) f(x) dx$$

## Examples of $h(x)$ :

1. To score Mean Free Path:
  - $h(x)$  = distance traveled.
2. To score Transmission:
  - $h(x) = \begin{cases} 1, & \text{if the particle crosses the region} \\ 0, & \text{if absorbed or lost.} \end{cases}$
3. To score Flux (Track Length Estimator):
  - $h(x) = \frac{L(x)}{V}$ , where  $L(x)$  is the length of the trajectory inside the volume  $V$

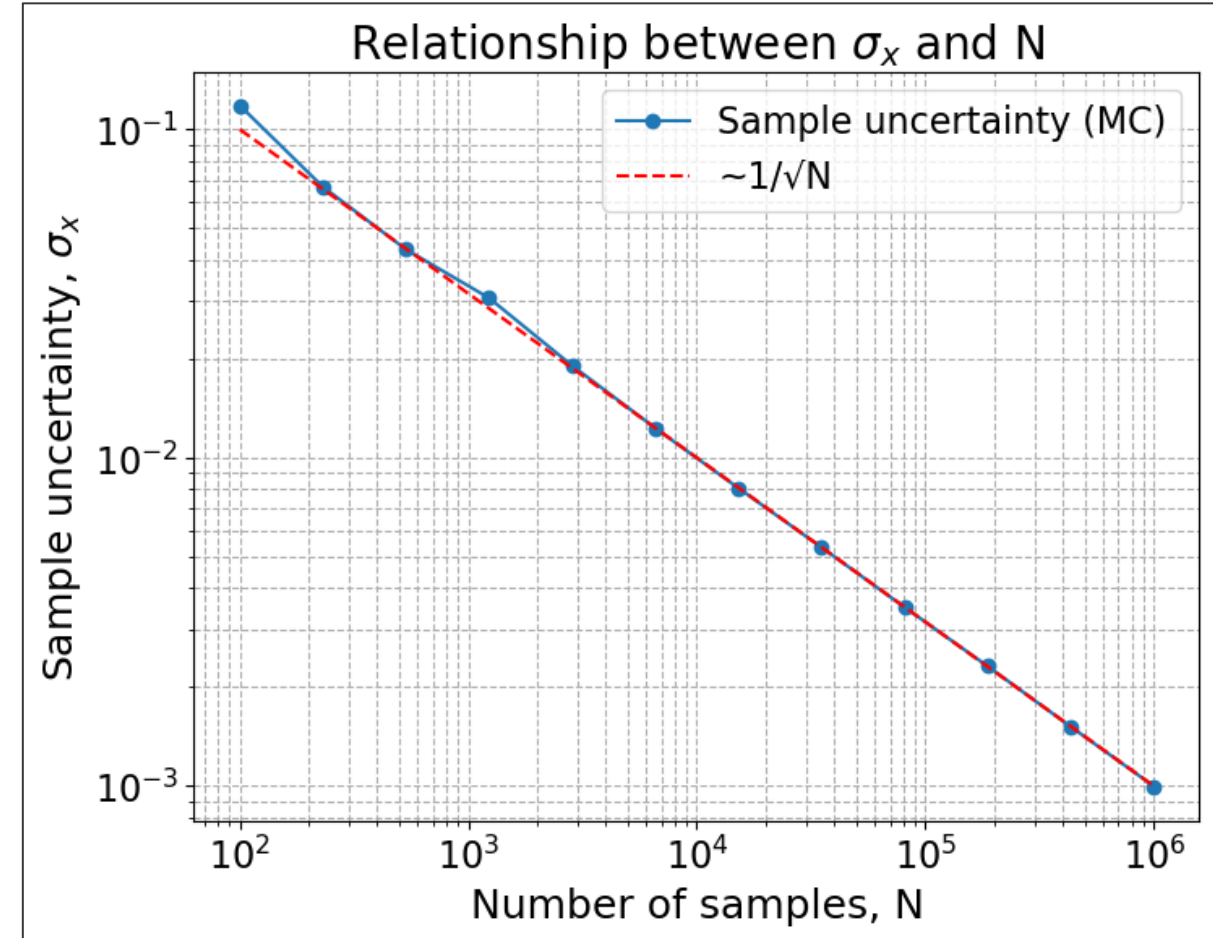
# Statistical Uncertainty Decreases as $1/\sqrt{N}$

The statistical error of the sample ( $\sigma_x$ ) depends on the number of histories ( $N$ ).

$$\sigma_x \propto \sqrt{\frac{1}{N}}$$

Slow convergence:

- To reduce uncertainty by 10x...
- **We need 100x more histories! (and 100x more CPU time)**

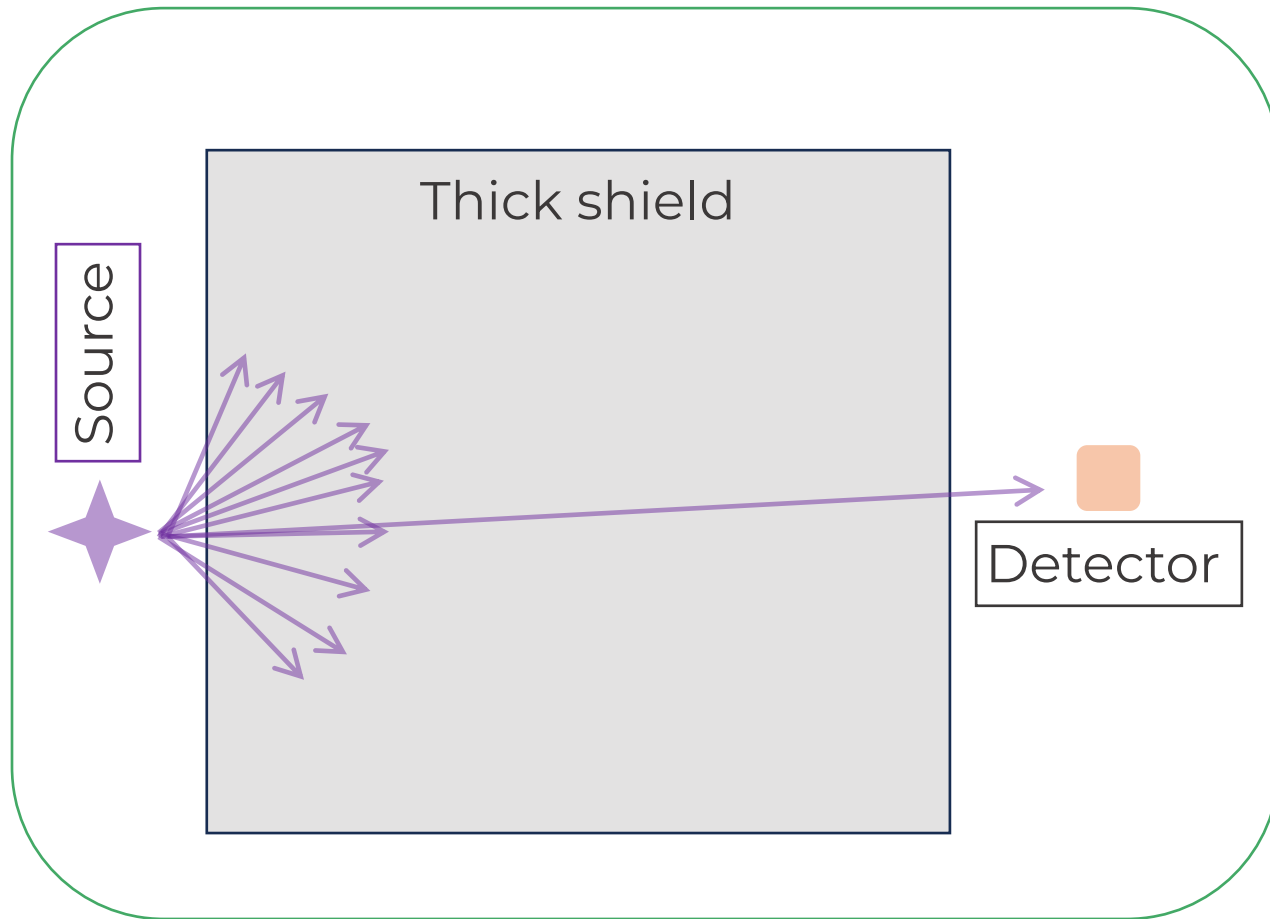


**Computing time grows quadratically with desired precision**

# Why (Sometimes) Brute Force Fails

# When particles have negligible contributions

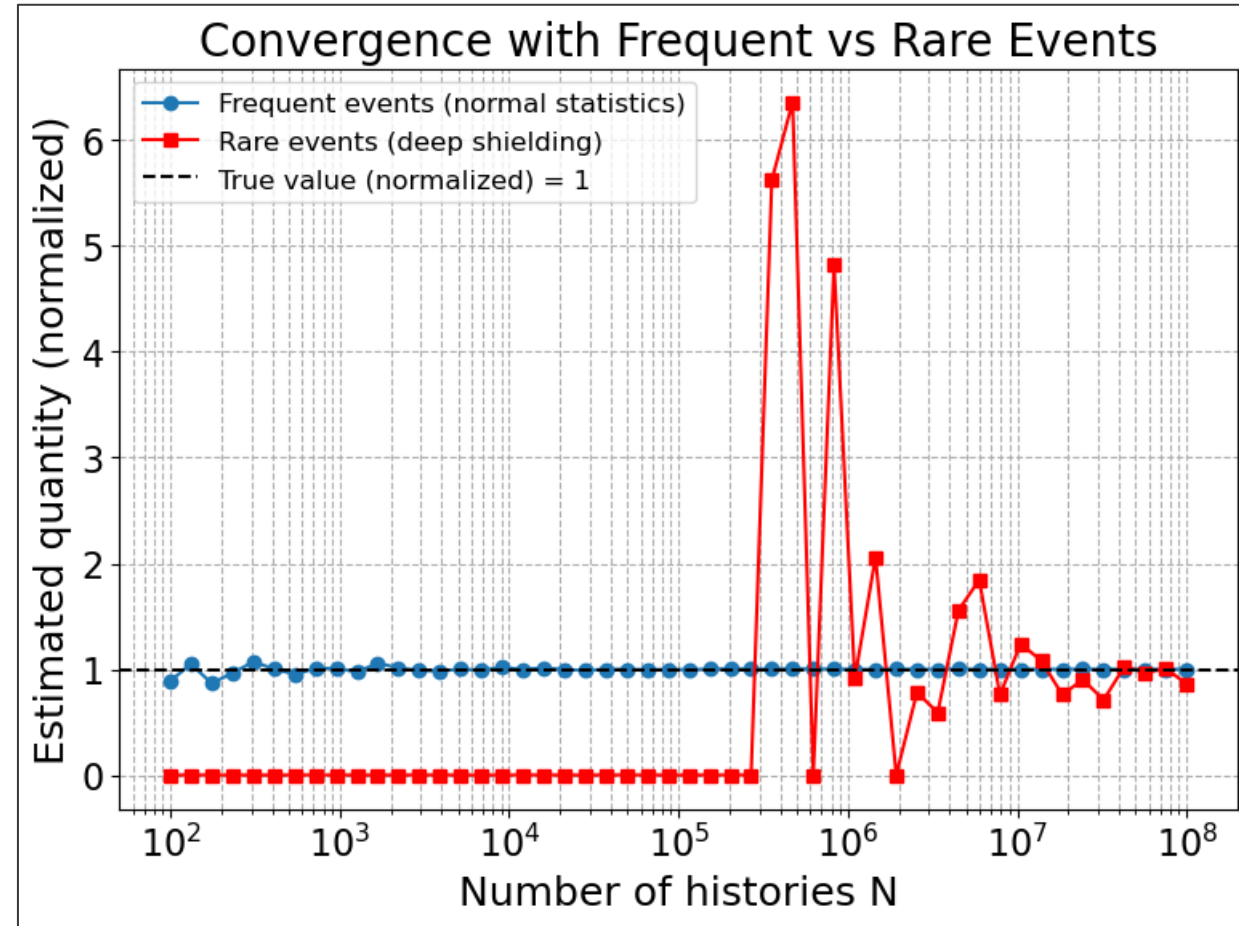
In some problems, most simulated particles **never contribute to the detector.**



- The Reality of Radiation Transport:
  - Particles scatter, absorb or leak out away from the detector.
- The “Zero” Contribution:
  - In shielding or collimator problems, most histories never reach the detector.
- CPU Implication:
  - We spend expensive computing time calculating detailed physics for particles that contribute nothing to the result.

# Rare events lead to high statistical variance

- The relative error  $R = \frac{\sigma_{\bar{x}}}{\bar{x}}$  tells us the **magnitude of the uncertainty compared to our result**.
- Rare events ( $p \ll 1$ ), **have huge variances compared to the mean**.
- The uncertainty is as big (or bigger) than the result itself.
- Translation: You need an astronomical N just to get a stable mean.





# Deep Shielding

- **The Physics of Shielding:**

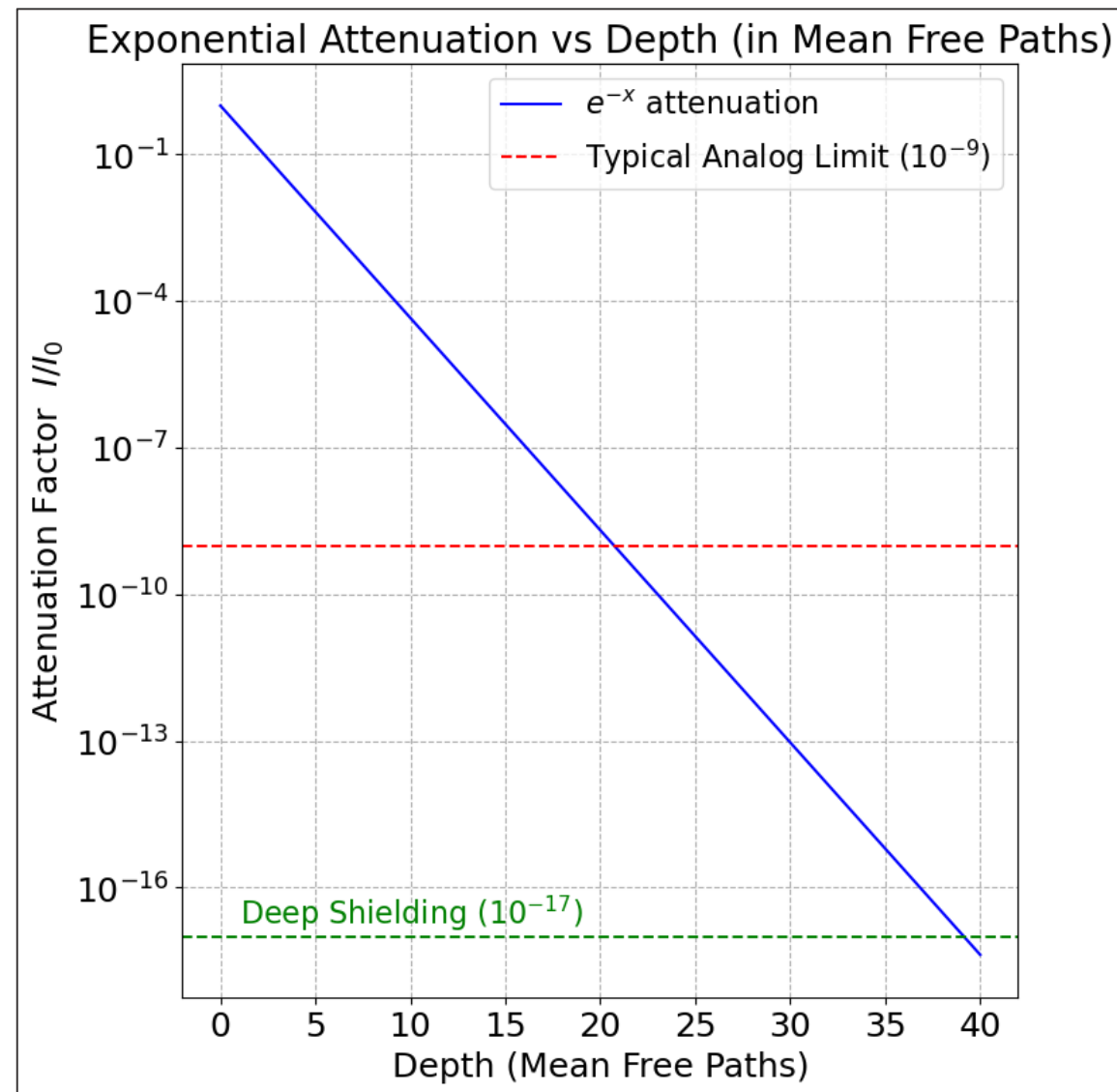
- Particles are removed exponentially:  $I(x) = I_0 e^{-\Sigma x}$

- **The magnitude:**

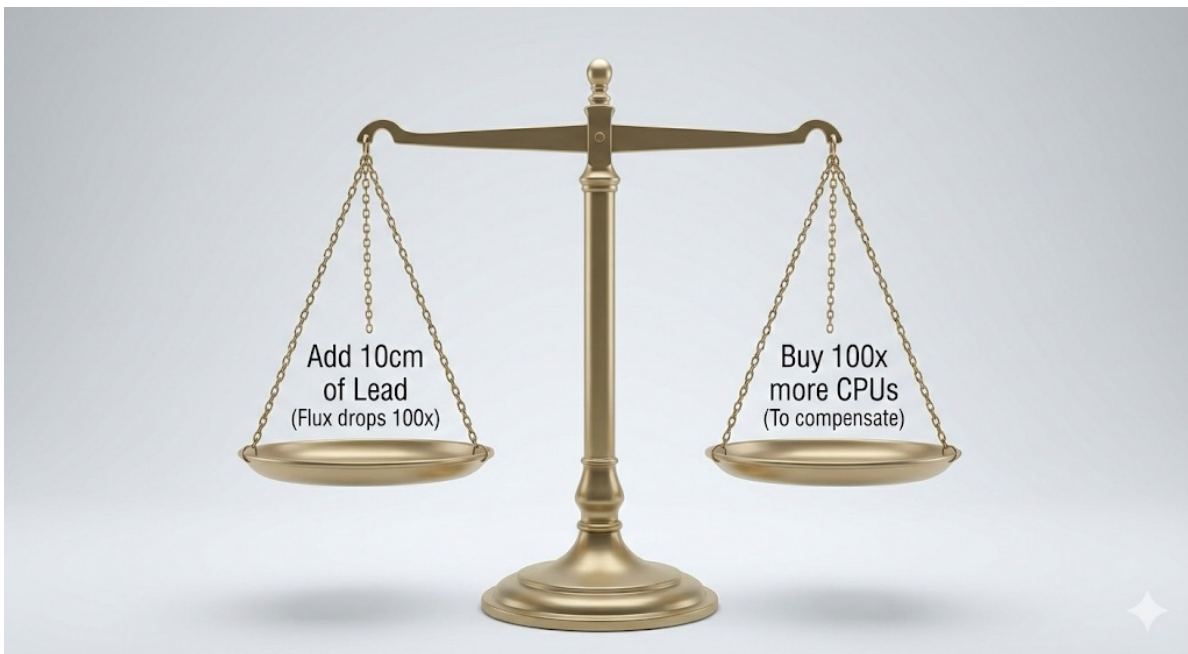
- To see **1 particle** through 10 Mean Free Paths (MFP):
  - Attenuation factor  $\propto e^{-10} \propto 4.5 \times 10^{-5}$
- Through 40 MFP (Deep Shielding):
  - Attenuation factor  $\propto 10^{-17}$

- **The implication:**

- We would need  $10^{17}$  histories just to get **one** count on average.



# Increasing N Alone Becomes Inefficient



Hardware just cannot keep up with exponential decay

## Moore's Law vs. Physics:

- Computers get faster (Linear/Polynomial Growth).
- Shielding difficulty grows **exponentially** with thickness.

## The scale:

- Simulating  $10^9$  histories: **Hours/Days** (Possible).
- Simulating  $10^{17}$  histories: **Centuries** (Impossible).

## The Verdict:

- We cannot solve deep penetration problems by adding more CPUs.
- **We must change the algorithm.**

# Variance and Efficiency

# Variance: Sample vs. Estimator

## The Sample Variance ( $\sigma^2$ )

- **What it is:** The spread of individual scores.
- **Where it comes from:** Physics of the Problem (PDF).
- **Behavior:** It converges to a constant value as  $N \rightarrow \infty$ 
  - It does NOT decrease with more histories.

The **Relative Error (R)** tells us the quality of our result:

$$R = \frac{\sigma_x}{\bar{x}} = \frac{\sigma}{\bar{x}\sqrt{N}}$$

## The Variance of the Mean ( $\sigma_x^2$ )

- **What it is:** The uncertainty of the estimator.
- **Where it comes from:** Depends on  $N$ .
- **The Link with the sample variance:**
$$\sigma_x^2 = \frac{\sigma^2}{N}$$
  - This is what we want to minimize

Rule of thumb:

| Relative Uncertainty | Quality of the score |
|----------------------|----------------------|
| 0.5 to 1             | Garbage              |
| 0.2 to 0.5           | Factor of a few      |
| 0.1 to 0.2           | Questionable         |
| < 0.1                | Generally reliable   |

# Efficiency: The Figure of Merit (FOM)

To measure the efficiency of a simulation, we define the **Figure of Merit**:

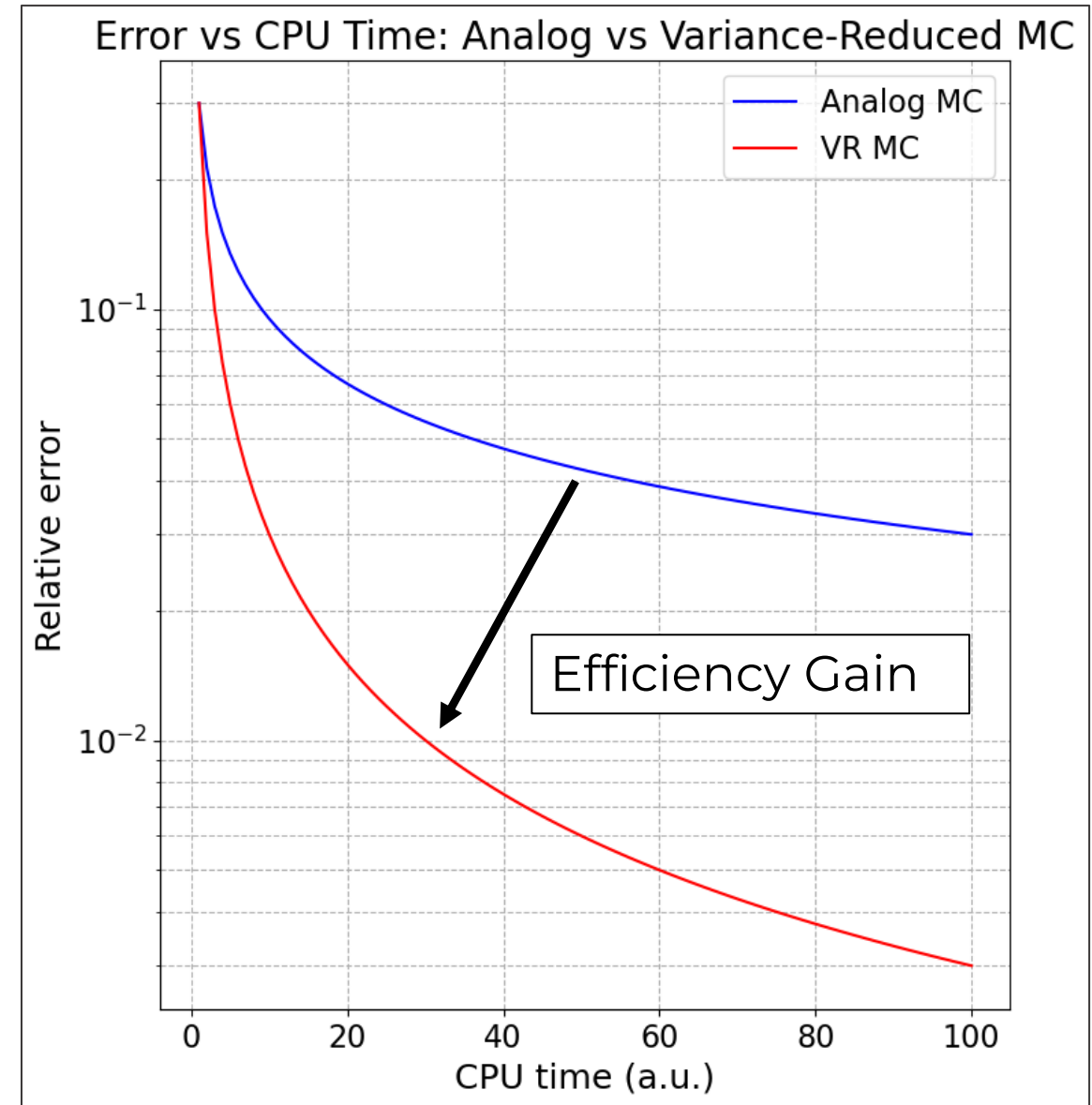
$$FOM = \frac{1}{\sigma_x^2 \times T}$$

- This metric takes into account the **variance and the CPU time**.
- Since  $\sigma_x^2 \propto \frac{1}{N}$  and  $T \propto N$ , then,  $FOM \approx \text{constant}$ .
- We cannot improve FOM by just running more histories.

A method is “better” only if the FOM increases.

# The Goal: Maximize the FOM

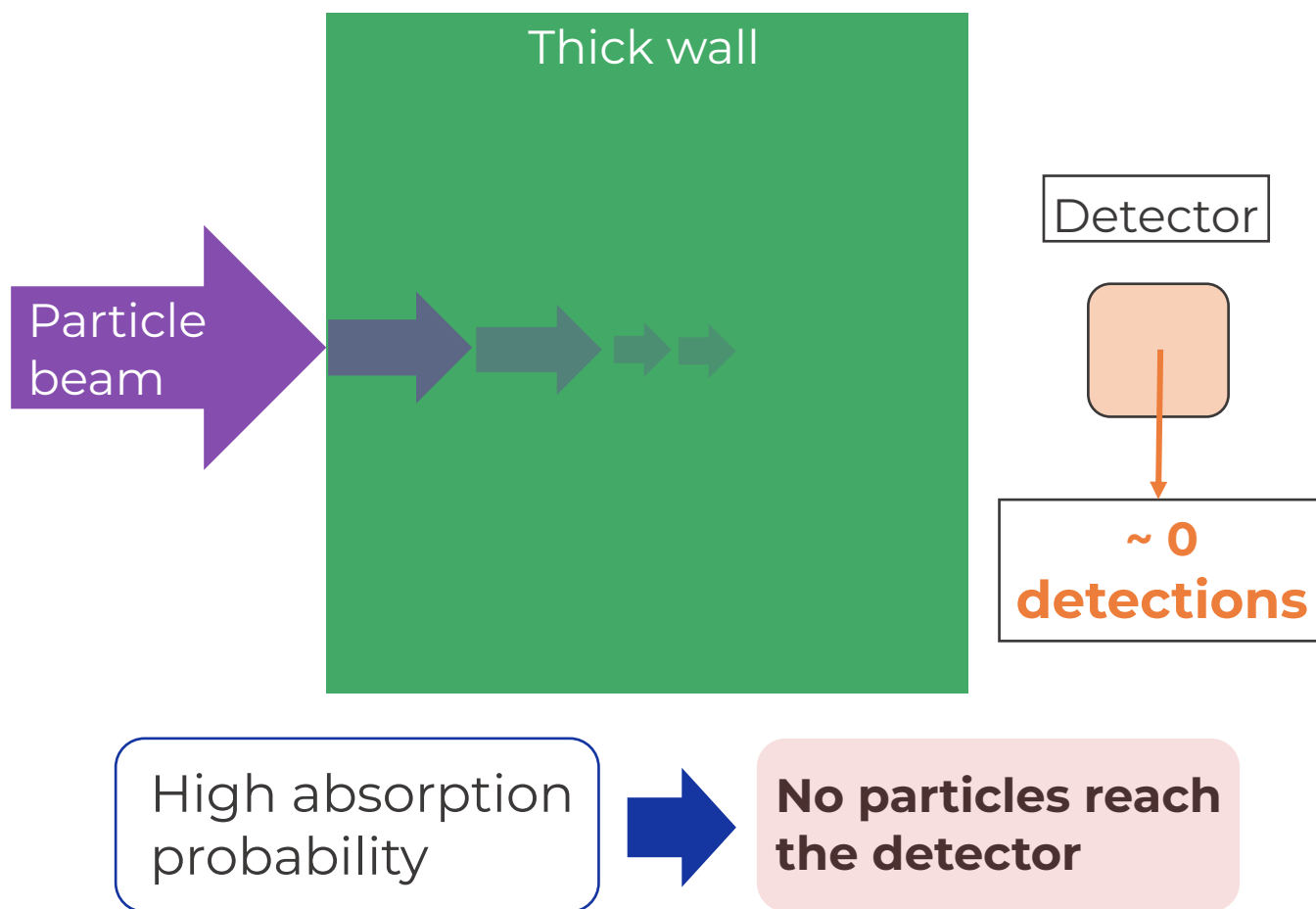
- What VR actually does:
  - **It modifies the sampling to reduce the Variance of the Expected Value.**
- The reduction in the variance must outweigh the increase in  $T$ .
- Two possible winning scenarios:
  1. **Same  $T$ : Get much smaller uncertainty**
  2. **Same Uncertainty: But in much lower  $T$ .**



# Motivation for Variance Reduction

# Analog sampling vs Biased sampling

In our previous example

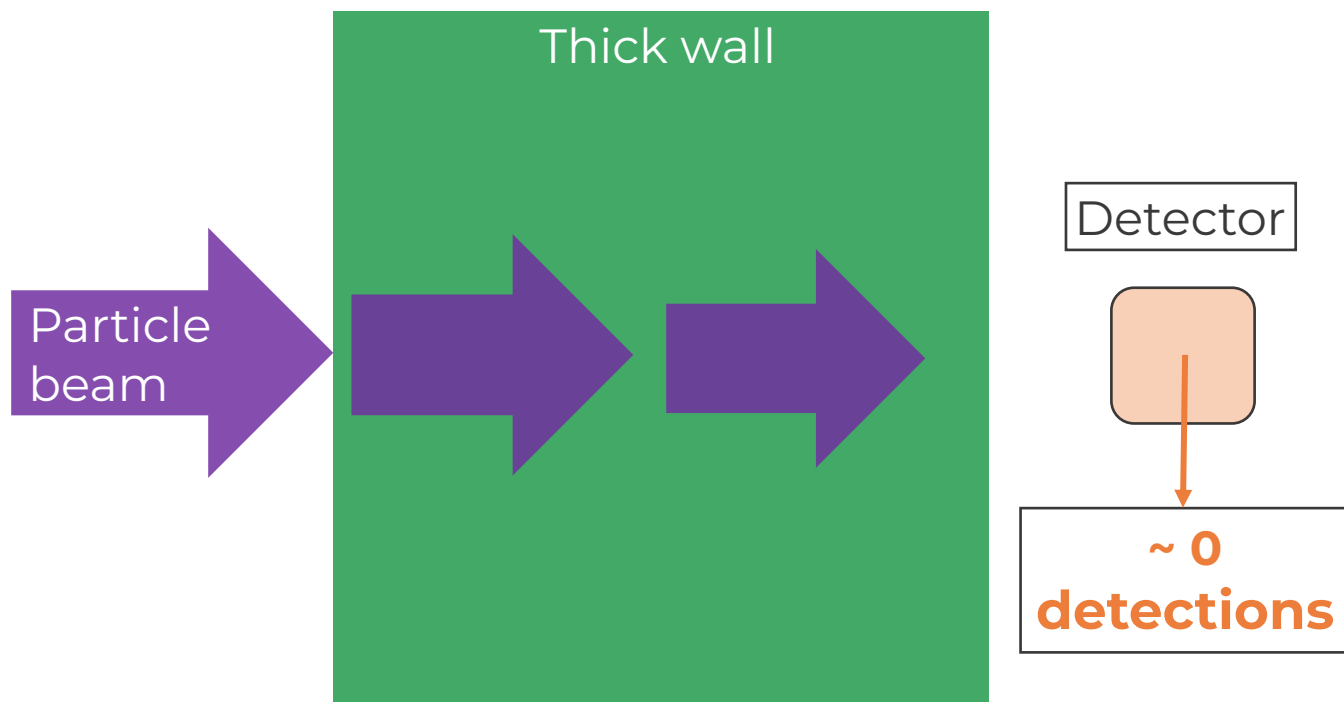


- Inside the shielding we are sampling from the **Physical PDF**  $f(x)$  (macroscopic absorption probability).
- **The result is:**
  - Correct physics, but,
  - Rare events are... well, rare.
- *Very inefficient for the case of shielding/deep penetration.*



# Analog sampling vs Biased sampling

But if we want more particles to reach the detector...



We sample from another PDF, **forcing the particles to penetrate the shield.**

- We force the simulation to sample from a **Biased PDF**  $\hat{f}(x)$ .
- This biased PDF should favour the outcome we want.
- Example: We force particles to penetrate the shielding (e.g. reducing the probability of absorptions).
- **The Problem:**
  - We are now simulating physics that do not correspond to reality.

# Preserving the mean

Remember: our goal in Monte Carlo is to estimate the mean value of an observable through repeated random sampling from the underlying probability distributions.

- Goal: Calculate the Physical Expectation Value  $E[h(x)]$
- We multiply and divide by the biased PDF  $\hat{f}(x)$ :
  - $E[h(x)] = \int h(x) \left[ \frac{f(x)}{\hat{f}(x)} \right] \hat{f}(x) dx$
- And define  $w = \frac{f(x)}{\hat{f}(x)}$  as the **Statistical Weight**.

- So, in a simulation:
  1. Sample  $x$  from  $\hat{f}(x)$  (Biased).
  2. Set statistical weight  $w = f(x)/\hat{f}(x)$ .
  3. Score contributions as  $h(x) \cdot w$

**The statistical weight allows us to preserve physical results**

# Families of Variance Reduction Techniques

# The VR Landscape

We can bias spatial transport, weight evolution, interaction physics or source characteristics.

## Spatial

Splitting

Russian Roulette

## Weight Control

Weight windows

Population control

## Physics based

Forced collisions

Exponential  
Transform

Branch biasing

## Source

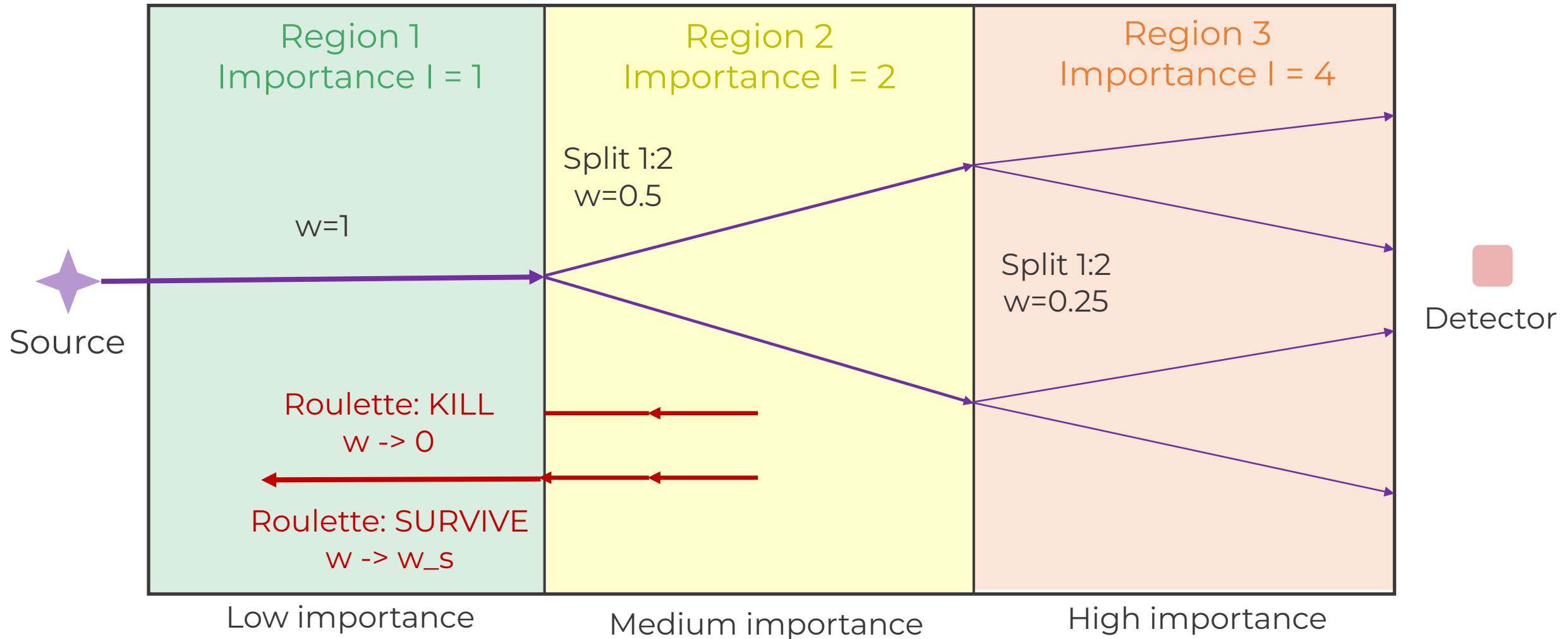
Bias initial energy

Bias initial  
direction

Which VR technique we choose will depend on the physics and goal of our simulation.

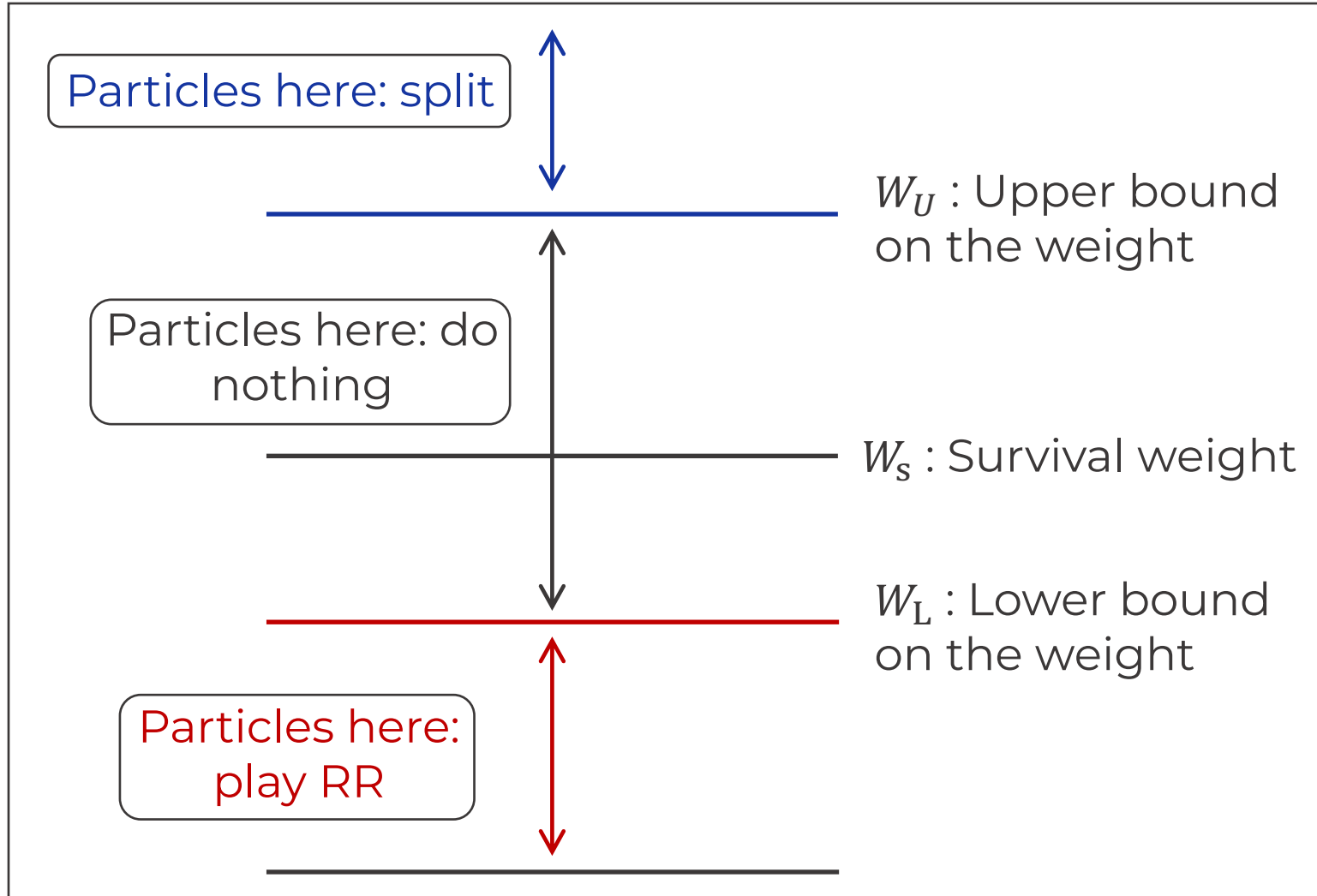
# Spatial Techniques

Control the number of particles across space based on importance



# Weight-Control Techniques

Splitting + RR based on space- and energy-dependent importance

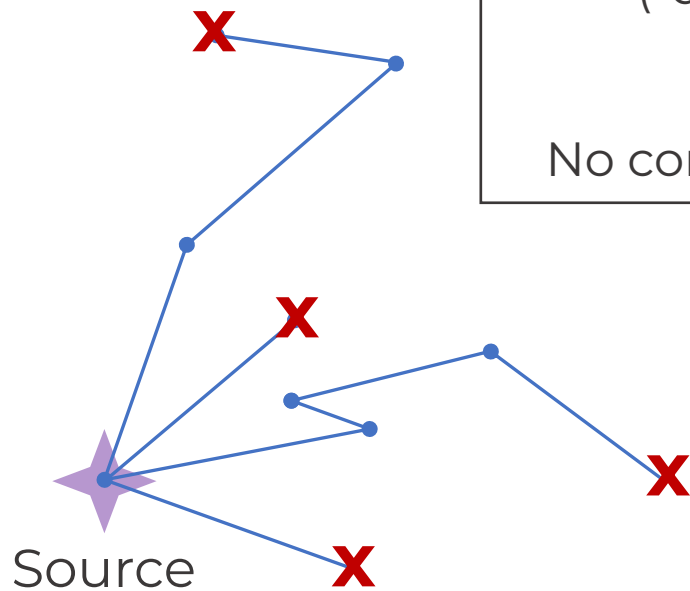


- To avoid "weight spikes"
- The idea is to keep the weights within a window or range.
- More details in next class!

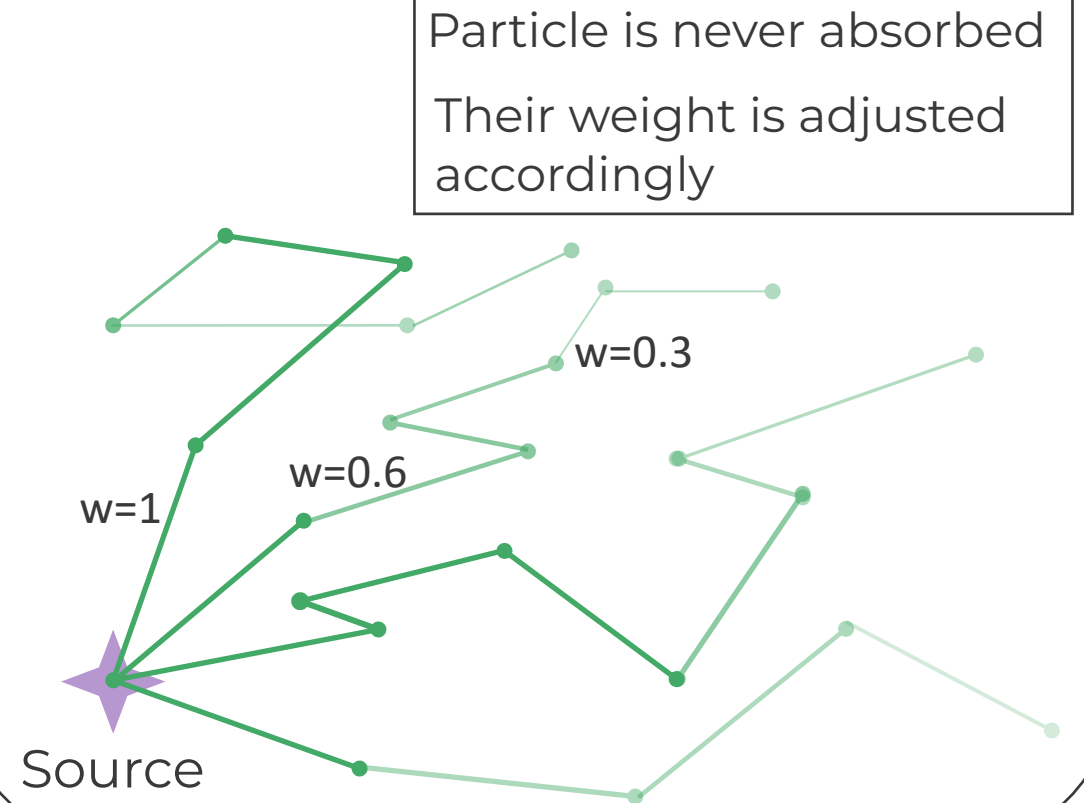
# Physics-Based Biasing

Increase probability of rare interaction channels or force interactions where relevant.

Analog Monte Carlo simulation



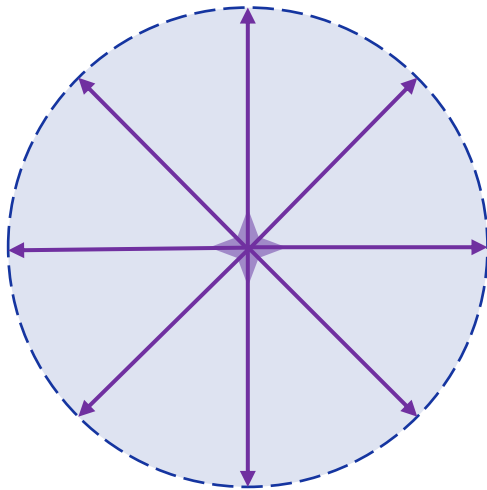
Non-Analog Monte Carlo simulation



# Source Biasing

Bias initial particle generation toward relevant energies or directions

Unbiased (analog) source

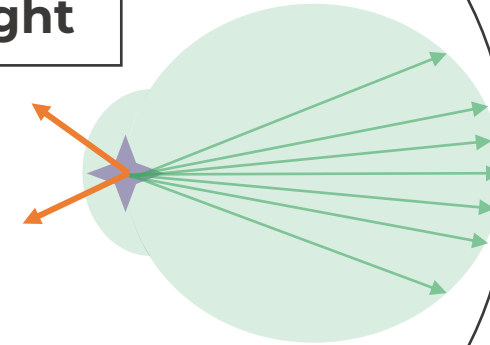


Emits in  $4\pi$ , most particles don't reach the detector

Biased (non-analog) source

Low Probability:  
**HIGH Weight**

High Probability:  
**LOW weight**



Biased emission, so more particles reach the detector



# Match each problem with its VR technique

Problem

Likely VR Approach

Deep shielding

Spatial + weight windows

Thin target physics

Physics biasing

Small detector/isotropic source

Source + spatial

Rare interaction channel

Physics biasing

# Takeaways

# Key Takeaways

- Monte Carlo simulates individual particle histories estimates physical observables.
- Statistical uncertainty decreases as  $1/N$ , but transport cost increases with  $N$ .
- Analog Monte Carlo is inefficient when contributions are rare.
- **Goal  $\neq$  only reduce error  $\rightarrow$  Goal = maximize efficiency of the simulation.**

# Biased Sampling -> Unbiased Physical Estimate?

**So, does Biased Sampling yield an Unbiased Estimate?**

**YES, providing the statistical weight is properly corrected.**

**Next Block: We'll see the details!**

